

Deriving the Harmonic Oscillator Ladder Operators from $x, p, \mathbf{x}, \mathbf{p}$ via the Commutator Action of H

We consider the one-dimensional quantum harmonic oscillator with Hamiltonian

$$H = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \quad (1)$$

and the canonical commutation relation

$$[\hat{x}, \hat{p}] = i\hbar. \quad (2)$$

We will treat the commutator

$$\text{ad}_H(\cdot) := [H, \cdot]$$

as a linear map acting on operators and show that suitable linear combinations of $\hat{x}$ and $\hat{p}$ become eigen-operators of ad_H . These eigen-operators will turn out to be the ladder operators a and $a^\dagger$.

Step 1: Computing $[H, \hat{x}][\mathbf{H}, \mathbf{x}]$ and $[H, \hat{p}][\mathbf{H}, \mathbf{p}]$

We use the standard identities

$$[AB, C] = A[B, C] + [A, C]B, \quad [A, BC] = [A, B]C + B[A, C]. \quad (3)$$

Commutator $[H, \hat{x}][\mathbf{H}, \mathbf{x}]$

We have

$$[H, \hat{x}] = \left[\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \hat{x} \right] \quad (4)$$

$$= \frac{1}{2m}[\hat{p}^2, \hat{x}] + \frac{1}{2}m\omega^2[\hat{x}^2, \hat{x}]. \quad (5)$$

Since $[\hat{x}^2, \hat{x}] = 0$, we only need

$$[\hat{p}^2, \hat{x}] = \hat{p}[\hat{p}, \hat{x}] + [\hat{p}, \hat{x}]\hat{p}. \quad (6)$$

Using $[\hat{p}, \hat{x}] = -[\hat{x}, \hat{p}] = -i\hbar$, we get

$$[\hat{p}^2, \hat{x}] = \hat{p}(-i\hbar) + (-i\hbar)\hat{p} = -2i\hbar\hat{p}. \quad (7)$$

Therefore,

$$[H, \hat{x}] = \frac{1}{2m}(-2i\hbar\hat{p}) = -\frac{i\hbar}{m}\hat{p}. \quad (8)$$

Result 1:

$$\boxed{[H, \hat{x}] = -\frac{i\hbar}{m}\hat{p}}. \quad (9)$$

Commutator $[H, \hat{p}][\mathbf{H}, \mathbf{p}]$

Similarly,

$$[H, \hat{p}] = \left[\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \hat{p} \right] \quad (10)$$

$$= \frac{1}{2m}[\hat{p}^2, \hat{p}] + \frac{1}{2}m\omega^2[\hat{x}^2, \hat{p}]. \quad (11)$$

We have $[\hat{p}^2, \hat{p}] = 0$, so only the second term remains:

$$[\hat{x}^2, \hat{p}] = \hat{x}[\hat{x}, \hat{p}] + [\hat{x}, \hat{p}]\hat{x} = \hat{x}(i\hbar) + (i\hbar)\hat{x} = 2i\hbar\hat{x}. \quad (12)$$

Hence,

$$[H, \hat{p}] = \frac{1}{2}m\omega^2(2i\hbar\hat{x}) = i\hbar m\omega^2\hat{x}. \quad (13)$$

Result 2:

$$\boxed{[H, \hat{p}] = i\hbar m\omega^2\hat{x}}. \quad (14)$$

Thus, the commutator action of H on the two-dimensional operator subspace $\text{span}\{\hat{x}, \hat{p}\}$ is linear:

$$[H, \hat{x}] = -\frac{i\hbar}{m}\hat{p}, \quad [H, \hat{p}] = i\hbar m\omega^2\hat{x}. \quad (15)$$

Step 2: Treating $[H, \cdot][\mathbf{H}, \cdot]$ as a Linear Map on Operator Space

Define the “vector” of operators

$$\vec{X} := \begin{pmatrix} \hat{x} \\ \hat{p} \end{pmatrix}. \quad (16)$$

Then we can write

$$\text{ad}_H(\hat{x}) := [H, \hat{x}] = -\frac{i\hbar}{m}\hat{p}, \quad (17)$$

$$\text{ad}_H(\hat{p}) := [H, \hat{p}] = i\hbar m\omega^2\hat{x}. \quad (18)$$

In matrix form,

$$\text{ad}_H \begin{pmatrix} \hat{x} \\ \hat{p} \end{pmatrix} = \begin{pmatrix} 0 & -\frac{i\hbar}{m} \\ i\hbar m\omega^2 & 0 \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{p} \end{pmatrix}. \quad (19)$$

Factoring out $i\hbar$, we define the real matrix

$$N := \begin{pmatrix} 0 & -\frac{1}{m} \\ m\omega^2 & 0 \end{pmatrix}, \quad (20)$$

so that

$$\text{ad}_H(\vec{X}) = i\hbar N \vec{X}. \quad (21)$$

Thus, on the operator subspace spanned by $\hat{x}, \hat{p}$, the map ad_H is represented by the 2×2 matrix $i\hbar N$.

Step 3: Solving the Eigen-Operator Problem $[H, O] = \lambda O$

We now look for an operator of the form

$$O = \alpha \hat{x} + \beta \hat{p}, \quad (22)$$

with complex coefficients α, β , such that

$$[H, O] = \lambda O, \quad (23)$$

i.e. O is an eigen-operator of ad_H .

Using the linearity of the commutator and the results above,

$$[H, O] = \alpha[H, \hat{x}] + \beta[H, \hat{p}] \quad (24)$$

$$= \alpha \left(-\frac{i\hbar}{m} \hat{p} \right) + \beta (i\hbar m \omega^2 \hat{x}). \quad (25)$$

Rearranging,

$$[H, O] = (i\hbar m \omega^2 \beta) \hat{x} + \left(-\frac{i\hbar}{m} \alpha \right) \hat{p}. \quad (26)$$

On the other hand,

$$\lambda O = \lambda(\alpha \hat{x} + \beta \hat{p}) = (\lambda \alpha) \hat{x} + (\lambda \beta) \hat{p}. \quad (27)$$

Since $\hat{x}$ and $\hat{p}$ are linearly independent as operators, we must have

$$\begin{cases} i\hbar m \omega^2 \beta = \lambda \alpha, \\ -\frac{i\hbar}{m} \alpha = \lambda \beta. \end{cases} \quad (28)$$

Solving for the Eigenvalues λ

From the second equation,

$$-\frac{i\hbar}{m} \alpha = \lambda \beta \quad \Rightarrow \quad \beta = -\frac{i\hbar}{m\lambda} \alpha. \quad (29)$$

Substitute into the first equation:

$$i\hbar m\omega^2 \left(-\frac{i\hbar}{m\lambda} \alpha \right) = \lambda \alpha. \quad (30)$$

Simplifying,

$$-i^2 \frac{\hbar^2 \omega^2}{\lambda} \alpha = \frac{\hbar^2 \omega^2}{\lambda} \alpha = \lambda \alpha. \quad (31)$$

Assuming $\alpha \neq 0$, we cancel α and obtain

$$\frac{\hbar^2 \omega^2}{\lambda} = \lambda \quad \Rightarrow \quad \lambda^2 = \hbar^2 \omega^2 \quad \Rightarrow \quad \lambda = \pm \hbar \omega. \quad (32)$$

Thus there are two eigen-operators $O_{\pm}$ satisfying

$$[H, O_{\pm}] = \lambda_{\pm} O_{\pm}, \quad \lambda_{\pm} = \pm \hbar \omega. \quad (33)$$

Finding the Corresponding Eigen-Operators

For $\lambda = -\hbar \omega$, we have

$$\beta_- = -\frac{i\hbar}{m(-\hbar\omega)} \alpha = \frac{i}{m\omega} \alpha, \quad (34)$$

so

$$O_- = \alpha \hat{x} + \beta_- \hat{p} = \alpha \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right). \quad (35)$$

For $\lambda = +\hbar \omega$, we have

$$\beta_+ = -\frac{i\hbar}{m(+\hbar\omega)} \alpha = -\frac{i}{m\omega} \alpha, \quad (36)$$

so

$$O_+ = \alpha \hat{x} + \beta_+ \hat{p} = \alpha \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right). \quad (37)$$

Up to an overall (non-zero) scalar factor, we therefore obtain the two special linear combinations

$$\boxed{O_- \propto \hat{x} + \frac{i}{m\omega} \hat{p}, \quad O_+ \propto \hat{x} - \frac{i}{m\omega} \hat{p}.} \quad (38)$$

They satisfy

$$\boxed{[H, O_-] = -\hbar \omega O_-, \quad [H, O_+] = +\hbar \omega O_+.} \quad (39)$$

These are precisely the eigen-operators of the adjoint action of H on the operator space spanned by $\hat{x}, \hat{p}$.

Step 4: Normalization and the Standard Ladder Operators

We now choose an overall normalization such that the commutator becomes $[a, a^\dagger] = 1$. Define

$$a = c \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right), \quad a^\dagger = c \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right), \quad (40)$$

with $c \in \mathbb{R}$ for simplicity, so that $a^\dagger$ is indeed the Hermitian adjoint of a . Using the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$, we compute

$$[a, a^\dagger] = c^2 \left[\hat{x} + \frac{i}{m\omega} \hat{p}, \hat{x} - \frac{i}{m\omega} \hat{p} \right] \quad (41)$$

$$= c^2 \left([\hat{x}, \hat{x}] - \frac{i}{m\omega} [\hat{x}, \hat{p}] + \frac{i}{m\omega} [\hat{p}, \hat{x}] - \frac{1}{m^2\omega^2} [\hat{p}, \hat{p}] \right). \quad (42)$$

We use

$$[\hat{x}, \hat{x}] = 0, \quad [\hat{p}, \hat{p}] = 0, \quad [\hat{x}, \hat{p}] = i\hbar, \quad [\hat{p}, \hat{x}] = -i\hbar, \quad (43)$$

so

$$[a, a^\dagger] = c^2 \left(-\frac{i}{m\omega} (i\hbar) + \frac{i}{m\omega} (-i\hbar) \right) \quad (44)$$

$$= c^2 \left(\frac{\hbar}{m\omega} + \frac{\hbar}{m\omega} \right) = c^2 \frac{2\hbar}{m\omega}. \quad (45)$$

We want $[a, a^\dagger] = 1$, therefore

$$c^2 \frac{2\hbar}{m\omega} = 1 \quad \Rightarrow \quad c^2 = \frac{m\omega}{2\hbar} \quad \Rightarrow \quad c = \sqrt{\frac{m\omega}{2\hbar}}. \quad (46)$$

Hence the normalized ladder operators are

$$a = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i}{\sqrt{2m\hbar\omega}} \hat{p}, \quad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} - \frac{i}{\sqrt{2m\hbar\omega}} \hat{p}. \quad (47)$$

They satisfy

$$[H, a] = -\hbar\omega a, \quad [H, a^\dagger] = +\hbar\omega a^\dagger, \quad [a, a^\dagger] = 1. \quad (48)$$

In summary, starting from the canonical pair $(\hat{x}, \hat{p})$ and viewing $\text{ad}_H(\cdot) = [H, \cdot]$ as a linear operator on the space of operators, we solved the eigen-operator equation $[H, O] = \lambda O$ and found that the eigen-operators are precisely the ladder operators a and $a^\dagger$, corresponding to eigenvalues $-\hbar\omega$ and $+\hbar\omega$, respectively.